

Here's the clean, complete system state document for Aletheia Fusion Intelligence:

Aletheia Fusion Intelligence (AFI) — Complete System State
Company: Aletheia | Product: Aletheia Fusion Intelligence
For new Claude threads — paste this at the start of every session
Last updated: 2026-03-23

Identity & Naming

Company: Aletheia
Product: Aletheia Fusion Intelligence (AFI)
Category: Incident Intelligence Platform
Tagline: Fewer Alerts. Real Incidents Only.
Federated net: Phoenix Signal Network (Genesis Layer)
Founded by: Tom Nelmes, Florida
AI advisors: Claude (Anthropic) — primary build partner
GPT — cross-validation, architectural review
Technical validator: Paul (senior Oracle engineer, ~28.5yr tenure)
Personal investment confirmed pending Oracle separation.
No written equity agreements while he remains W2.

Core Thesis

Everything is signals. Every digital system produces behavioral signals as a byproduct of existing. AFI's fusion engine detects behavioral anomalies in signal streams — not cyber attacks specifically.

The connector library determines which signals enter the engine.
The detector packs determine which behavioral geometries are recognised.
The fusion math underneath is domain-agnostic by architecture.

The engine does not care what signals it fuses. It cares about behavioral geometry — the shape of how things move, timing, sequence, and structural signature of normal versus anomalous. This is not a metaphor. It is literally how the codebase is architected.

Location

~/sentinel-v2

Key commands:

Start server: node src/index.js

Handshake: ~/sentinel-v2/handshake.sh

Short version: ~/sentinel-v2/handshake.sh --short

Tech Stack

- Node.js v18.19.1 + Express
- TOML config (@iarna/toml)
- Winston logging
- Nodemailer (email alerts via Mailtrap)
- uuid@8
- better-sqlite3 (alert persistence — Phase 15a)
- EventBus (custom pub/sub, internal)
- Docker + docker-compose (Phase 11)

Architecture

Event-driven pipeline. All components communicate via EventBus.

Channels:

event:ingested	Raw event received via POST /ingest
event:normalized	After normalizer transforms to internal schema
detector:finding	Each detector publishes its score here
detector:error	Detector failures
fusion:complete	After fusion engine scores all detectors
alert:triggered	Alert fired to all channels
containment:queued	Containment actions queued for human approval
system:kill_switch	Emergency stop

Pipeline flow:

POST /ingest

→ Normalizer

→ Engine (runs all 12 detectors in parallel)

→ Fusion (weighted scoring + gate logic)

→ Alert (console + email + slack + webhook)

→ Containment (ip_block, email_hold, token_revoke, quarantine)

→ Audit log (immutable, append-only)

→ Alert persisted to SQLite (data/alerts.db)

Fusion Formula

$S = (\sum w_i \times s_i) / (\sum w_i + \epsilon)$ where $\epsilon = 0.0001$

$S_{\text{final}} = \max(S_{\text{base}}, \text{gate_floor})$ if any gate condition triggers

Severity thresholds:

LOW < 0.40

MEDIUM 0.40 – 0.60

HIGH 0.60 – 0.80

CRITICAL ≥ 0.80

AI layer annotates only, never suppresses.

The deterministic fusion engine is inviolable.

Detectors — Tier 1 (all ENABLED)

Name	Weight	Gate	Notes
meta_fusion	1.4	0.80	Fires if ≥ 3 detectors active AND policy_violation OR privilege_gradient present. Gate floor 0.80.
policy_violation	1.3	0.60	Any policy breach → gate floor 0.60
privilege_gradient	1.3	0.60	Any escalation → gate floor 0.60
tempo_cadence	1.2	none	Rate anomaly detection
beaconing	1.2	none	C2 interval pattern (CV thresholds tightened Phase 11 to reduce noise)
fan_out	1.1	none	Lateral spread detection
sequence_chaining	1.1	none	Multi-step attack chain detection
dormant_asset	1.0	0.60	age ≥ 30 d + sensitive + novelty
drift_detection	1.0	none	Behavioral drift from baseline
first_seen	0.9	none	New IP/device/relationship
baseline_confidence	0.9	none	Low baseline confidence penalty
identity_volatility	0.8	none	Rapid identity attribute changes

Detectors — Tier 2 (DISABLED — baseline-dependent)

sensitive_asset_spread 1.2 Broad sensitive resource access

slow_accumulation 1.1 Low-and-slow privilege accumulation
cross_identity_sync 1.2 Coordinated multi-identity behavior

Detectors — Tier 3 (DISABLED — enterprise-grade, graph memory required)

entropy_deviation 1.3
graph_topology 1.3
provenance_trust 1.2
context_collapse 1.2
multi_node_drift 1.1

File Structure

```
~/sentinel-v2/
├── config/
│   └── sentinel.toml
├── data/
│   ├── alerts.db      ← SQLite alert store (Phase 15a)
│   └── baselines.json ← Baseline persistence
├── src/
│   ├── index.js      ← Express API, EventBus wiring
│   ├── core/
│   │   ├── config.js
│   │   ├── eventBus.js
│   │   ├── normalizer.js
│   │   ├── fusion.js
│   │   ├── baseline.js
│   │   └── engine.js
│   ├── utils/
│   │   ├── logger.js
│   │   └── auditLog.js
│   └── modules/
│       ├── detectors/
│       │   ├── BaseDetector.js
│       │   ├── metaFusion.js
│       │   ├── policyViolation.js
│       │   ├── privilegeGradient.js
│       │   ├── tempoCadence.js
│       │   ├── beaconing.js
│       │   ├── fanOut.js
│       │   ├── sequenceChaining.js
│       │   └── dormantAsset.js
```

- driftDetection.js
 - firstSeen.js
 - baselineConfidence.js
 - identityVolatility.js
- alerts/
 - BaseAlert.js
 - alertStore.js ← SQLite-backed (Phase 15a)
 - consoleAlert.js
 - slackAlert.js ← disabled
 - emailAlert.js ← Mailtrap working
 - webhookAlert.js ← disabled
- containment/
 - BaseContainment.js
 - ipBlock.js
 - emailHold.js
 - tokenRevoke.js
 - quarantine.js
- integrations/
 - BaseConnector.js
 - firewallConnector.js
 - identityConnector.js
 - edrConnector.js
 - webhookConnector.js
 - m365Connector.js ← Phase 13
 - rmmConnector.js ← Phase 13
- governance/
 - killSwitch.js
 - alertRateLimit.js
 - configValidator.js
 - explainability.js
 - evidenceReport.js ← Phase 14
- metrics/
 - metricsCollector.js
- scripts/
 - ingest-poller.js ← Phase 13 — polls M365 + RMM on schedule
 - approval-queue.js
 - stress-test-100.js
- logs/
 - sentinel.log
 - audit.log
- dashboard.html ← Phase 10 operator dashboard
- Dockerfile ← Phase 11
- docker-compose.yml ← Phase 11
- .env

└─ handshake.sh

API Endpoints

GET /health
GET /status
POST /ingest body: event object
GET /containment/pending
POST /containment/approve/:actionId body: {approved_by, action_type}
POST /containment/deny/:actionId body: {denied_by, action_type}
GET /metrics
GET /baselines
GET /alerts
POST /alerts/:alertId/acknowledge body: {operator}
POST /alerts/:alertId/suppress body: {operator, duration_seconds}
POST /alerts/:alertId/notes body: {operator, note}
POST /governance/killswitch body: {action, operator}
GET /report/:alertId ← Phase 14 — Evidence Report
GET /audit-chain ← Phase 14 — full audit chain
GET /dashboard ← Phase 10 — operator UI

Auth: X-Sentinel-Key header required on all endpoints except /health and /dashboard

Alert Channels

console ☒ working
email ☒ working — Mailtrap SMTP
slack ☐ disabled — needs webhook_url in sentinel.toml
webhook ☐ disabled — needs url in sentinel.toml

Containment Actions (all human-approval gated)

ip_block queues → pending_approval → approve/deny
email_hold queues → pending_approval → approve/deny
token_revoke queues → pending_approval → approve/deny
quarantine queues → pending_approval → approve/deny

Real API integrations: simulation mode until MSP pilot provides credentials.

Integration Connectors (Phase 12 + 13)

Firewall connector	simulation mode
Identity connector	simulation mode (Azure AD, Okta, LDAP)
EDR connector	simulation mode (CrowdStrike, Defender, SentinelOne)
Webhook connector	simulation mode (Splunk, ServiceNow, PagerDuty, Slack)
M365 connector	simulation mode — fetches signin + audit logs via Graph API
RMM connector	simulation mode — ConnectWise, NinjaRMM, Datto

⚠ Phase 12a TABLED — live connector validation requires real MSP environment.
Simulation mode complete. Resumes at first pilot deployment.

Ingest poller: scripts/ingest-poller.js
Polls M365 + RMM on configurable interval (M365_POLL_INTERVAL_SECONDS)
Posts normalized events to /ingest automatically
Both connectors report simulation mode until .env credentials configured

Evidence Reports (Phase 14)

GET /report/:alertId returns:

- report_id, generated_at, generated_by
- Incident header (alert_id, event_id, severity, detected_at, status)
- Fusion summary (final_score, top_drivers, deterministic note)
- Detector findings (all 12 detectors, scores, weights, rationale)
- Human decisions (operator, timestamp, action)
- Containment actions (approved/denied, who, when)
- Full audit chain (all entries for this event)
- Compliance note (SOC 2, HIPAA, ISO 27001, CMMC, cyber insurance)

GET /audit-chain?limit=N&type=TYPE returns filtered audit log entries.

Alert Persistence (Phase 15a)

Alerts stored in SQLite: data/alerts.db
Survive server restarts. Loaded automatically on startup.
Fields: alert_id, event_id, tenant_id, severity, top_drivers,
final_score, timestamp, status, acknowledged_by,

acknowledged_at, suppressed_until, suppressed_by, notes

⚠ Phase 15b TABLED — Redis + horizontal scaling.
Trigger: 50k+ events/day through single instance OR 200+ MSP
client environments feeding one deployment. File and revisit.

Audit Log Record Types

AUDIT_INIT
EVENT_INGESTED
DETECTOR_FINDINGS
FUSION_RESULT
ALERT_FIRED
CONTAINMENT_ACTION
GOVERNANCE_*
ERROR

Git History (phase tags)

phase-6b-complete	adversarial testing — 20 events, 6 attack classes, 72ms, 0 false negatives
phase-7-complete	API hardening — X-Sentinel-Key, rate limiting, HMAC
phase-8-complete	Tier 2 detectors (disabled by default)
phase-9-complete	Tier 3 detectors (disabled by default)
phase-10-complete	Operator dashboard — live alerts, containment UI, metrics
phase-11-complete	Dockerfile, docker-compose, graceful shutdown, env profiles
phase-12-complete	Integration framework — firewall, identity, EDR, webhook connectors
phase-13-complete	M365 + RMM connectors, ingest-poller, simulation mode
phase-14-complete	Evidence reports, audit chain endpoint, compliance output
phase-15a-complete	SQLite alert persistence, alerts survive restarts

Validated Pipeline Test Results

[meta_fusion]	score=0	hint=low	(correct — scores others, not self)
[policy_violation]	score=0.9	hint=critical	
[privilege_gradient]	score=0.9	hint=critical	
[tempo_cadence]	score=0.5	hint=medium	
[beaconing]	score=0.9	hint=critical	
[fan_out]	score=0	hint=low	

[sequence_chaining] score=0 hint=low
[dormant_asset] score=0.75 hint=high
[drift_detection] score=0.05 hint=low
[first_seen] score=0.4 hint=medium
[baseline_confidence] score=0.2 hint=low
[identity_volatility] score=0.2 hint=low

Fusion: base=0.42 gate=0.80 final=0.80 severity=CRITICAL ☒

Alert: console ☒ email ☒ slack ☐ webhook ☐

Containment: ip_block ☒ email_hold ☒ token_revoke ☒ quarantine ☒

Evidence report: 12 findings, full audit chain ☒

Alert persistence: survives restart ☒

Standard Test Event

```
curl -s -X POST http://localhost:3000/ingest \
```

```
-H "Content-Type: application/json" \
```

```
-H "X-Sentinel-Key: change-me-before-production" \
```

```
-d
```

```
'{"event_type":"login_attempt","category":"auth","source":{"ip":"10.0.0.1","identity":"user_42"},"structural":{"rate_per_min":25,"privilege":{"current":1,"requested":5,"delta":4},"first_seen":{"ip_for_identity":true,"device_for_identity":true},"dormancy_days":45,"action_type":"admin_login"},"observations":{"window_seconds":40,"counts":{"events":25,"targets":17,"failures":23},"intervals_ms":[1000,1020,980,1010,990,1005]}'
```

Expected: severity=CRITICAL, score=0.80,
drivers=policy_violation/privilege_gradient/beaconing

Master Pack Library

Pack	Ship Phase	Detectors	Priority	Regulatory Angle
MSP Operations Pack	NOW	6	Critical	Cross-tenant credential isolation
Healthcare Pack (HIPAA)	Phase 1	8	High	HIPAA §164.312, PHI access audit
Financial Pack	Phase 1	8	High	SOX, PCI-DSS, GLBA, wire fraud
Cloud Security Pack	Phase 1-2	10	High	CSA CCM, AWS/Azure, CSPM gap-fill
Identity Risk Pack	Phase 1-2	6	High	Zero Trust, NIST 800-207
Legal Pack	Phase 2	6	Medium	Matter confidentiality, ABA duties

Email & Collaboration Pack	Phase 2	8	Medium	Email DLP, M365, OAuth governance
Endpoint Pack	Phase 2	7	Medium	EDR augmentation, MITRE ATT&CK
Compliance Pack	Phase 2	6	Medium	SOC 2, ISO 27001, CMMC
OT / Industrial Pack	Phase 3	8	Medium	NERC CIP, IEC 62443, CISA ICS
Insurance Underwriting Pack	Phase 3	7	Medium	Carrier requirements, risk scoring
Static Systems / Physical	Phase 3	6	Medium	Physical-digital convergence
Supply Chain Pack	Phase 3-4	7	Medium	NIST SP 800-161, vendor risk
Telecommunications Pack	Phase 4	7	Medium	FCC, STIR/SHAKEN, fraud detection
AI Agent Security Pack	Phase 4	8	High	AI governance, agentic oversight
Robotics & Autonomous Pack	Phase 4-5	7	Strategic	OT/robotics standards
Defense & Intelligence Pack	Phase 5	8	Strategic	CMMC L3, NIST 800-171, DoD supply

Total detector target: 100+ across all packs

MSP Operations Pack ships first — protects the MSP's own business (cross-tenant credential abuse, RMM propagation attacks).

First sales conversation: "this protects YOUR business" not "your clients."

Phoenix Signal Network — Genesis Layer

Federated, opt-in, privacy-preserving signal intelligence layer.

Shares behavioral signal abstractions across all AFI deployments.

Signals only — never semantic data, never raw events.

The compounding moat: no competitor can replicate historical signal data or cross-deployment learning from a standing start.

Status: Architectural intent. Not yet implemented.

Build phase: Post-Phase 15. Requires multiple live deployments first.

Remaining Build Phases

CURRENT STATE

Detection pipeline 

Fusion scoring 

Alert delivery  (console + email)

Containment queue  (human-approval gated)

Audit log	✓ (immutable, structured)
API hardening	✓ (X-Sentinel-Key, rate limiting)
Tier 2 detectors	✓ (built, disabled by default)
Tier 3 detectors	✓ (built, disabled by default)
Operator dashboard	✓ (live alerts, containment, metrics)
Docker / deployment	✓
Integration framework	✓ (simulation mode)
M365 + RMM connectors	✓ (simulation mode, live via .env)
Ingest poller	✓ (polls on schedule, feeds pipeline)
Evidence reports	✓ (/report/:alertId, /audit-chain)
Alert persistence	✓ (SQLite, survives restart)

Next Build Priorities

MSP Operations Pack — first detector pack, ships before IT Nation

Detectors:

cross_tenant_credential	Cross-tenant credential reuse
admin_login_multi_client	Admin login across multiple clients
shared_credential_pattern	Shared credential detection
rmm_command_propagation	Unusual RMM command propagation
msp_lateral_movement	Lateral movement across client tenants
after_hours_admin	After-hours admin activity anomaly

Healthcare Pack (HIPAA) — Phase 1 after MSP pack

Dashboard improvements — evidence report view in UI

Handshake script update — rename Phoenix Sentinel references to AFI

Tabled

Phase 12a — live connector validation

Trigger: first MSP pilot provides credentials

Phase 15b — Redis, horizontal scaling

Trigger: 50k+ events/day OR 200+ client environments

Long Horizon (file, don't chase)

Phoenix Signal Network build-out (post live deployments)

Defense/military contracting (separate entity, software layer)

Novel monetary system concept

Robotics + AI agent applications of fusion engine

Go-To-Market

Primary channel: MSPs (Managed Service Providers)

Positioning: Completing existing stacks, not competing

Sits above detection layer, below analyst

"Make your SIEM useful" not "replace your SIEM"

Pricing model: \$300–500/environment/month (SMB)

\$500–900/environment/month (MSP target)

\$2,000+/environment/month (enterprise/gov)

MSP Operations Pack: included for all MSP partners

Billback model: MSP pays AFI \$500/env → bills client \$900-1,200/env

MSP margin: \$400–700/environment/month

Pilot offer: 60–90 day free pilot for first MSP

Deployment time: ~2 hours per client environment

Ongoing support: ~30–60 min/month per environment at steady state

Revenue milestones:

50 MSPs × 60 clients × \$500 = \$1.5M ARR

1,000 MSPs = ~\$232M ARR (current pricing)

Target conference: IT Nation Connect — Orlando — November 4–6, 2026

Conference goal: 5 real follow-up conversations (not 5 sales)

Attend as: Networking attendee, not exhibitor

Business Formation

Company: Aletheia (Florida LLC)

Domain: AletheiaFI.com (Cloudflare ~\$10-12/yr)

Email: Tom@AletheiaFI.com (Microsoft 365 Business Basic \$6/mo)

Banking: Mercury (no fees)

Trademark: USPTO Class 042 — attorney review before public use

Formation status: In progress

Investor Context

Two external investors: serious interest, pre-revenue, staged tranches
Paul: personal investment confirmed, pending Oracle separation
Commitment: original pricing terms honored permanently regardless of valuation
No written equity agreements with Paul while W2 at Oracle

Equity program for early MSPs:
0.05% equivalent per partner, first 50–100 MSPs
Attorney engagement required before implementation
Warrants or SAFEs preferred over direct equity

Performance guarantee concept:
\$10,000–\$50,000 remedy if signal reduction thresholds not met
180–360 day window
Legal counsel required before any public statement

Key Design Principles

1. AI layer annotates only, never suppresses.
Deterministic core is inviolable.
2. Incident object is the system of record — not just an alert.
Carries timeline, entities, detector findings, fusion math,
analyst notes, approvals, containment actions, audit chain.
3. Completing, not competing.
AFI fills the incident intelligence gap above existing stacks.
Microsoft Sentinel leaves this gap even in E5 environments.
4. MSP structural pain with Microsoft Sentinel:
Per-workspace billing, per-tenant rule maintenance,
cross-workspace query caps (5–20 workspaces).
AFI is architecturally designed for MSP scale.
5. Pricing as insurance:
\$300–500/environment/month < 0.5% of smallest client's annual revenue.

How To Start A New Session

1. Run handshake: ~/sentinel-v2/handshake.sh

2. Paste this document + handshake output to new Claude thread
3. Add human note:
 - Goal right now:
 - What I just changed:
 - What's broken (if anything):
 - What "done" looks like today:
4. Start building

Save that as AFI_SYSTEM_STATE.md in the project root and commit it. That's the canonical document going forward — every new thread starts with this plus the handshake output. Good session Tom. Phases 13 through 15a in one run. The build is pilot-ready.